HAUTE ÉCOLE D'INGÉNIERIE ET DE GESTION DU CANTON DE VAUD

BACHELOR THESIS

RAW: a GitLab Integration within Moodle

Department

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Date

22 August 2026

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Context

LMS and VCS platforms are both essential in computer science education, but operate independently, leading to fragmented workflows.

Teachers

- Create repositories manually
- Configure repository permissions by hand
- Manage multiple platforms

Students

- Inconsistent workflows between courses
- Switch between LMS and VCS platforms
- Grades, deadlines, repositories in different places

Platforms Involved

Learning Management System (LMS)

- Centralizes course organization and documents
- Manages assignments, submissions, and grading
- Structured teaching environment

Version Control System (VCS)

- Manages source code and development workflows
- Enables collaboration through Git repositories
- Provides project management / DevOps tools

Moodle

Modular, open-source, institutionally established

GitLab

Self-hosting, data sovereignty

Specification

Objectives

- Assess existing solutions and APIs
- Design a modular architecture
- Implement, deploy, test and validate

Expected Result

- Extensible Moodle-GitLab integration
- Automated, centralized workflows
- Simple, open-source solution

Deliverables

- Report and planning
- Presentation and poster
- Plugins, documentation, tests, and infrastructure

Design Principles

Design a solution around 3 core principles

Automation

- Automate GitLab resource management
- Favor flexibility over rigid workflows

LMS-native

- Provide a unified workflow
- Central access point over fragmented platforms

Extensible Design

- Open-source allowing contributions
- Modular architecture for futur extensions

State of the Art

	Automation	Open source	LMS integration	VCS independent
GitHub Classroom	✅	🟡	❌	❌
Codio	✅	❌	🟡	❌
Classroom 50	✅	✅	❌	❌
RepoBee	🟡	✅	❌	✅
Classmoji	✅	✅	✅	🟡
RAW	✅	✅	✅	🟡

Repository Assignment Workflow (RAW)

A Moodle-native, extensible, and open-source solution.

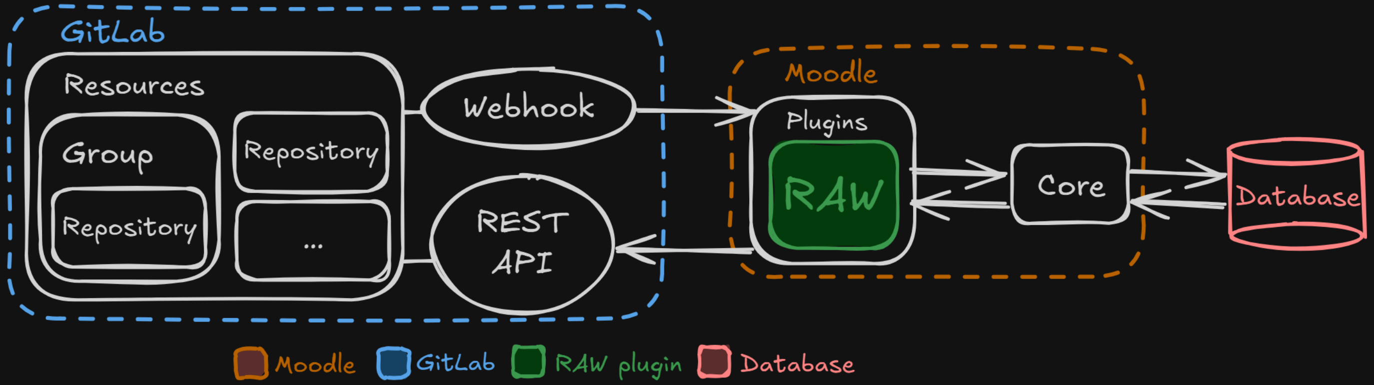

Figure 1. RAW architecture diagram

- Follows the 3 design principles defined previously
- Built using two Moodle plugins: **Custom Field**, **Activity Module**

GitLab Client

Abstraction layer between Moodle and GitLab REST API

Moodle
Activity Plugin → Entry Point
GitLab Client → Resource Clients
Groups, Projects, Users, Webhooks... → GitLab
REST API

```php
$client = new GitLabClient($token);
$client->group()->create(...);
$client->project()->fork(...);
```

- Abstracts HTTP requests behind a structured, intuitive interface
- Client mirrors GitLab's API resources structure
- Inspired by the **GitLab PHP API Client** library

GitLab Tokens

Secure persistence of sensitive values with encryption

1
Configuration
A teacher provides one GitLab token per Moodle course.

→

2
Encryption
Built-in Moodle encryption using `XSalsa20` and `Poly1305`.

→

3
Encrypted Storage
Only ciphertext is stored preventing token exposure from a database-only breach.

→

4
API Use
Moodle decrypts the token to authenticate GitLab API requests.

Webhooks

GitLab-to-Moodle synchronization through signed events

GitLab
Push, issue, or merge request event

→

Signed POST
Cryptographic signature

→

Moodle Endpoint
`/mod/gitlab/webhook.php`

→

Verified
Dispatch event

Per-Activity Secret

Random 32-byte key, stored encrypted in database.

Integrity Check

Computed signature compared with the received signature.

Replay Protection

Timestamps older than five minutes are rejected.

Only verified events can update templates, repositories, or submission status.

Demonstration

Features

Assignment Management

- RAW activity configuration
  - Due dates
  - Group capacity
  - Reviewers list
- Moodle calendar events and notifications
- Release solution access

User Workflows

- Teacher dashboard
  - Manage groups and members
  - Access template repository
  - Retrieve submissions
- Student interface
  - Create or join groups
  - Access repository information

Features

GitLab Integration

- Automated repository provisioning
- Repository access and permission management
- Webhook synchronization
- Protected files verification
- Custom GitLab host support

Limitations and Future Work

Limitations

- Limited by GitLab’s permission model
- Administrator access for plugin installation
- Manual Moodle–GitLab identity mapping

Future Work

- Improve error handling and user feedback
- Process long-running operations asynchronously
- Support additional Version Control Systems (VCS)
- Validate with representative users

Contributions

- Introduces a flexible, automated workflow
- Provides an extensible, open-source, Moodle-native integration
- Reduces manual coordination between LMS and VCS platforms
- Improves consistency and efficiency for instructors and students